

CSE370 Database Systems: Lab Assignment 2

Objective	Practice advanced SQL operations using a single table in MySQL.		
Due Date:	1 Week, submit before next class	Number of Questions:	10

Scenario:

You are working as a database administrator for a growing tech company. The HR department needs you to analyze employee data to generate various reports about salaries, departments, and organizational structure. All operations will be performed on a single table called **employees**.

Part 1: Understanding the Data

You will use a database called **Company_<Your8DigitStudentID>**.

Example:

```
CREATE DATABASE Company_12345678;  
USE Company_12345678;
```

The **employees** table has the following attributes and data types.

Table Name: employees	
Attribute Name	Attribute type
e_id	char(4)
first_name	varchar(20)
last_name	varchar(20)
hire_date	date
job_id	char(10)
salary	int
commission_pct	decimal(5,3)
manager_id	char(4)
department_id	char(3)

The HR department has collected the following employee data:

e_id	first_name	last_name	hire_date	job_id	salary	commission_pct	manager_id	dept_id
E001	John	Doe	2020-01-15	IT_PROG	65000	NULL	E010	D01
E002	Jane	Smith	2019-03-22	SA_REP	55000	0.150	E011	D02
E003	Michael	Johnson	2021-06-10	IT_PROG	62000	NULL	E010	D01
E004	Emily	Davis	2018-11-05	SA_REP	58000	0.200	E011	D02
E005	David	Wilson	2022-02-14	IT_PROG	60000	NULL	E010	D01
E006	Sarah	Brown	2020-07-30	HR_REP	50000	0.150	E012	D03
E007	James	Taylor	2019-09-18	SA_REP	60000	0.180	E011	D02
E008	Lisa	Anderson	2021-04-25	FI_ACC	52000	NULL	E013	D04
E009	Robert	Martinez	2020-12-01	IT_PROG	67000	NULL	E010	D01
E010	Patricia	Garcia	2017-05-10	IT_MAN	85000	NULL	NULL	D01
E011	Thomas	Lee	2016-08-20	SA_MAN	80000	0.100	NULL	D02
E012	Maria	White	2018-02-15	HR_MAN	75000	NULL	NULL	D03
E013	Daniel	Harris	2019-10-30	FI_MAN	78000	NULL	NULL	D04
E014	Jessica	Clark	2021-01-20	HR_REP	48000	0.100	E012	D03
E015	Kevin	Lewis	2022-03-12	FI_ACC	51000	NULL	E013	D04

Note: **commission_pct** is the commission percentage (for sales representatives only). **manager_id** references another employee who is the manager (NULL means the employee has no manager). **dept_id** indicates which department the employee belongs to.

Part 2: Questions

IMPORTANT: Provide one single query per question! Your query should not be hard coded, it should provide the logic mentioned in the question.

Write SQL queries to complete the following tasks: [10 * 1 = 10]

- Q1.** Create the employees table with appropriate data types for all columns shown above.
 - Q2.** Insert all the given data displayed in the table above into the employees table.
 - Q3.** Find the first_name, last_name, hire_date, and department_id of all employees with the latest hire_date.
 - Q4.** Find the first_name, last_name, e_id, salary, and department_id of the employee with the highest salary in each department.
 - Q5.** Find the first_name, last_name, e_id, commission_pct, and department_id of all employees in department 'D02' who have a higher commission_pct than all employees in department 'D01'.
 - Q6.** Find the department_id and total number of employees of each department that does not have any employee with a salary greater than 70000.
 - Q7.** For each department, find the department_id, job_id, and salary where the salary is less than at least one other job_id in that same department.
 - Q8.** Find the manager_id of managers who do not have any employee under them with a salary less than 55000.
 - Q9.** Find the first_name, last_name, e_id, salary, department_id, and commission_pct of the employee with the highest commission_pct under each manager.
 - Q10.** Find the job_id and average salary of job positions where the average salary is less than the overall average salary of all employees.
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Submission Instructions

Make a copy of the **submission template** in your own Google Drive. For each task in Part 2:

- 1. Write the MySQL command in your submission document.
 - 2. Execute the command in XAMPP shell (MySQL command line), take a screenshot showing the command and output, then paste it below the command in your document.
 - 3. Repeat for all 10 tasks.
 - 4. Download your document as PDF.
 - 5. Rename using format: **ID_Name_CSE370_Lab_Sec##_Assignment02_Summer26.pdf**
Example: **12345678_JohnDoe_CSE370_Lab_Sec12_Assignment02_Summer26.pdf**
 - 6. Submit via the Google Form in your Discord section channel.
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